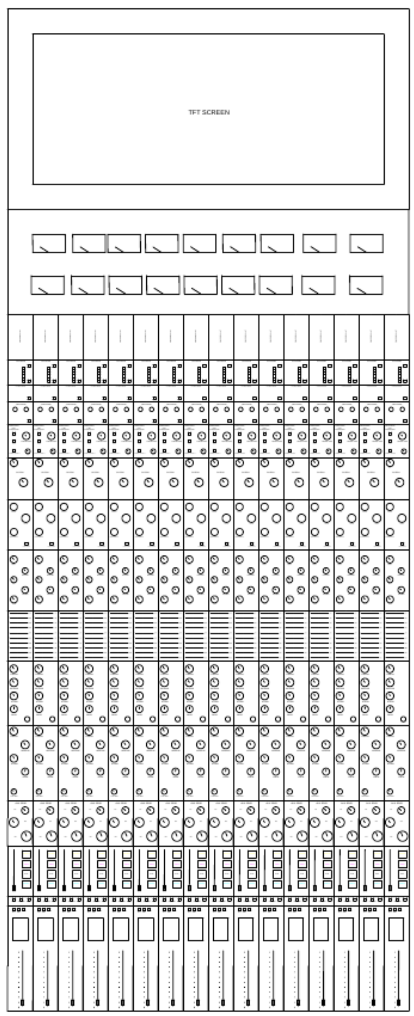
PRISM Hybrid Vision

Prototype 1A Specifications



Overview

The PRISM Hybrid Vision is a hybrid digital-analog immersive sound mixing and mastering console. The console is designed for DAW workflows and professional studio use. The console also has a modular architecture, so you can replace modules with new ones without replacing the entire console. The console also has full DAW automation without compromising on true analog sound. The console also has switchable preamp, conversion, and output styles. It can select from OLD, OLD-NEW, and NEW. Each channel has a 5 band and 10 band EQ, the 5 band is for broader selection of frequencies, the 10 band has a more sophisticated selection of frequencies to choose from. There are two compressors per channel. One VCA and one Opto compressor chained together. These two compressors create the best duo per channel for excellent sound. Each channel comes with 3 Stereo AUX sends. Each channel also comes with MUTE, SOLO, RECORD, and CUT buttons, which are fully automated from the center console.

What Problem Does This Console Solve?

- No compromises: This console combines true analog warmth with digital control, automation, and DAW integration.
- Instead of buying different types of processors, you can select them purely on the console.
- Digital routing and processing makes studio workflow faster using automation and recall.
- A modular architecture with a modular interface lets you integrate new equipment with the universal connector.

Key Features

- Hybrid analog paths
- Digital recall
- Motorized faders
- Modular Cards
- Switchable preamp color and style
- 2 Compressors per channel
- Optical Networking
- Touchscreen control
- Expandable architecture

General Specifications

Specification	Target
Input Channels	48 (Expandable)
Mic Preamps	48
Line inputs	48
Stereo Returns	16
AUX Sends	16
Sub-Groups	8
Outputs	Stereo + Monitor
Sample Rates	44.1-192 kHz
Bit Depth	48-24 Bit
Digital Interface	Optical + USB

Automation	Full Recall
Faders	100 mm Motorized
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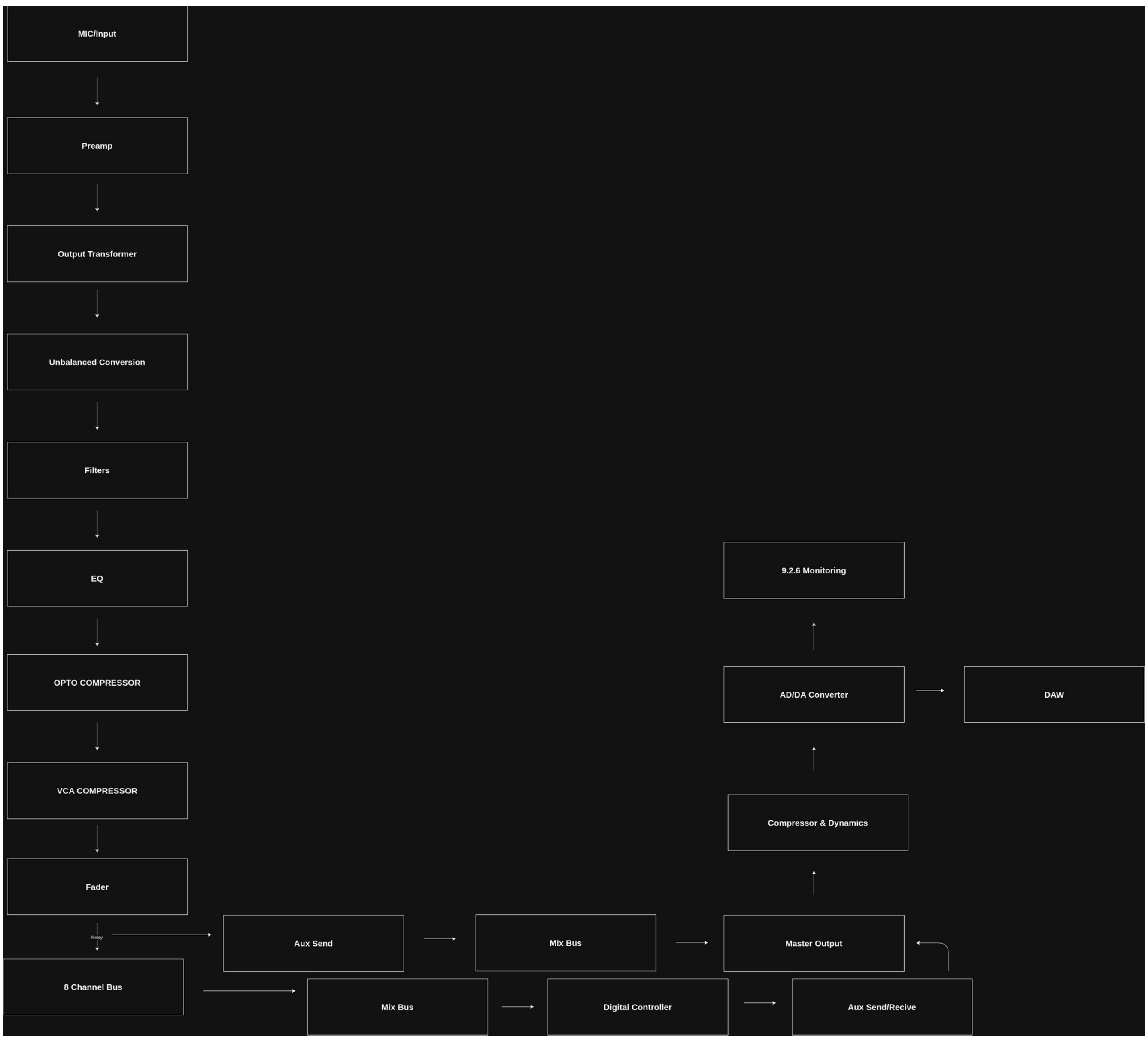
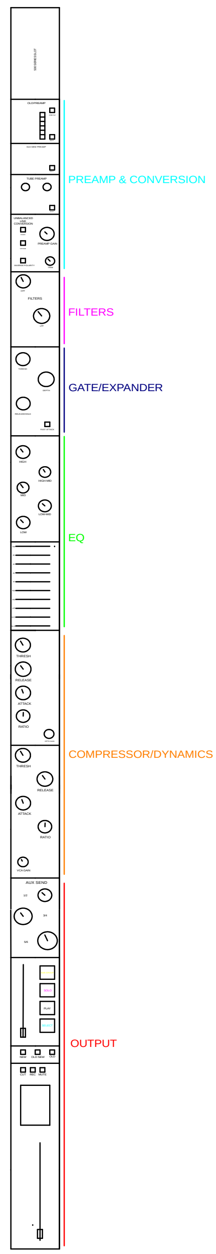
System Diagram



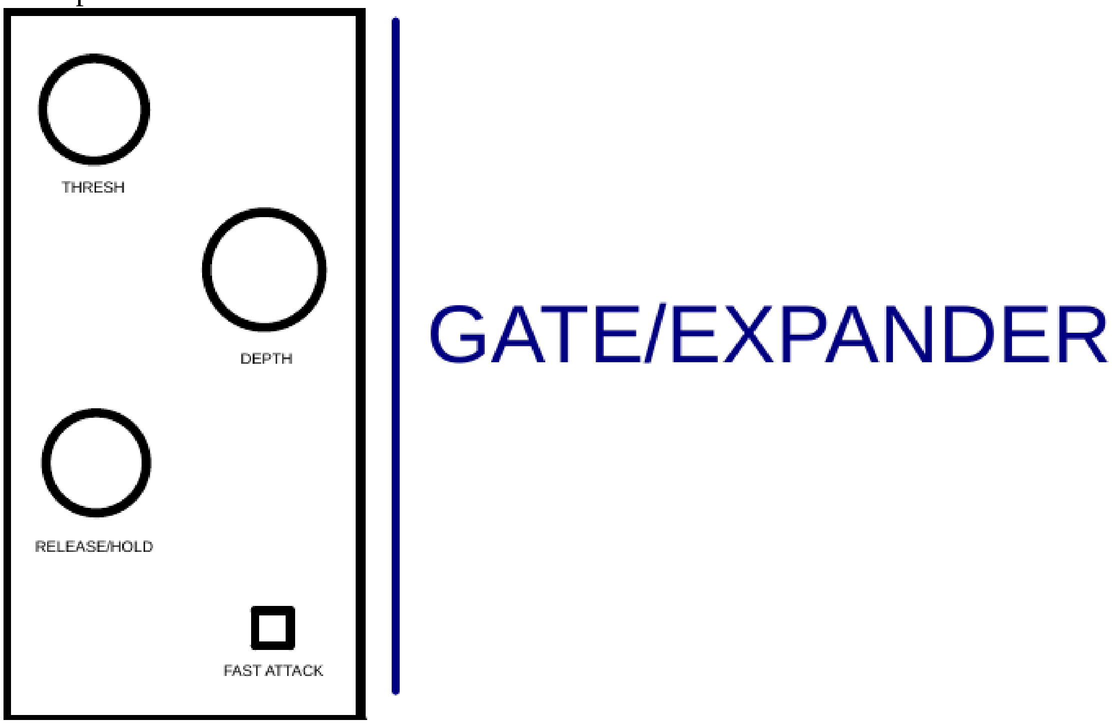

PREAMP & CONVERSION
FILTERS
GATE/EXPANDER
EQ
AUX SEND
COMPRESSOR/DYNAMICS
OUTPUT

Mic Preamplifier

- Gain Range: +34dB to +64dB
- -20dB Pad button
- EIN: -129 dBu
- Phantom Power: +48V DC (Switchable)
- Input Impedance: 1500 Ω Nominal
- Maximum Input Level: +8dBu (No pad) / +28 dBu (With Pad)
- Frequency Response: 30 Hz to 20 kHz
- THD+N: Less than 0.05% at +4dBu output

Dynamic Processing

Gate/Expander



- Analog Noise Gate/Expander: Active VCA sidechain comparator network driving a THAT 2180 VCA chip. Sweepable Threshold, Depth (0 to -40dB clamping), and a Release (50ms to 3s). Includes a Fast Attack switch for sub-100 millisecond tracking.

Compressors

- Two-stage Cascaded Compressor Engine:
    - Harmonic Opto Compressor: Utilizes a analog optical vactrol element (VTL5C4) for broad, musical gain reduction and vintage low-end program-dependent saturation characteristics. Includes makeup gain.
    - VCA Compressor: Utilizes the THAT 4320 integrated dynamics engine for blistering, lightning-fast transient peak capping (6.1 mV per decibel log-scaling gate).

Equalizer Module

- Module 1: 5-band sweepable Parametric EQ
  - Architecture: Active multi-stage variable filter network utilizing dial potentiometers for frequency tracking per band.
  - Frequency Coverage: Full audio spectrum coverage split across 5 continuous overlapping operational bands.
    - Low Band: Sweeps 20 Hz to 200 Hz.
    - Low-Mid Band: Sweeps 100 Hz to 1 kHz.
    - Mid Band: Sweeps 500 Hz to 5 kHz.
    - High-Mid Band: Sweeps 1 kHz to 10 kHz.
    - High Band: Sweeps 5kHz to 20 kHz.
- Module 2: 10-Band Graphic Equalizer
  - Architecture: High-density proportional-Q active bandpass filter matrix driven by 10 physical front-panel vertical slider potentiometers.
  - Fixed ISO Frequencies: 31.25 Hz, 62.5 Hz, 125 Hz, 250 Hz, 500 Hz, 1 kHz, 2 kHz, 4 kHz, 8 kHz, and 16 kHz.

Modular Processing

Interchangeable modules:

- Preamp Module
- Dynamic processing modules
- EQ Module
- AUX Send Module
- Fader Module

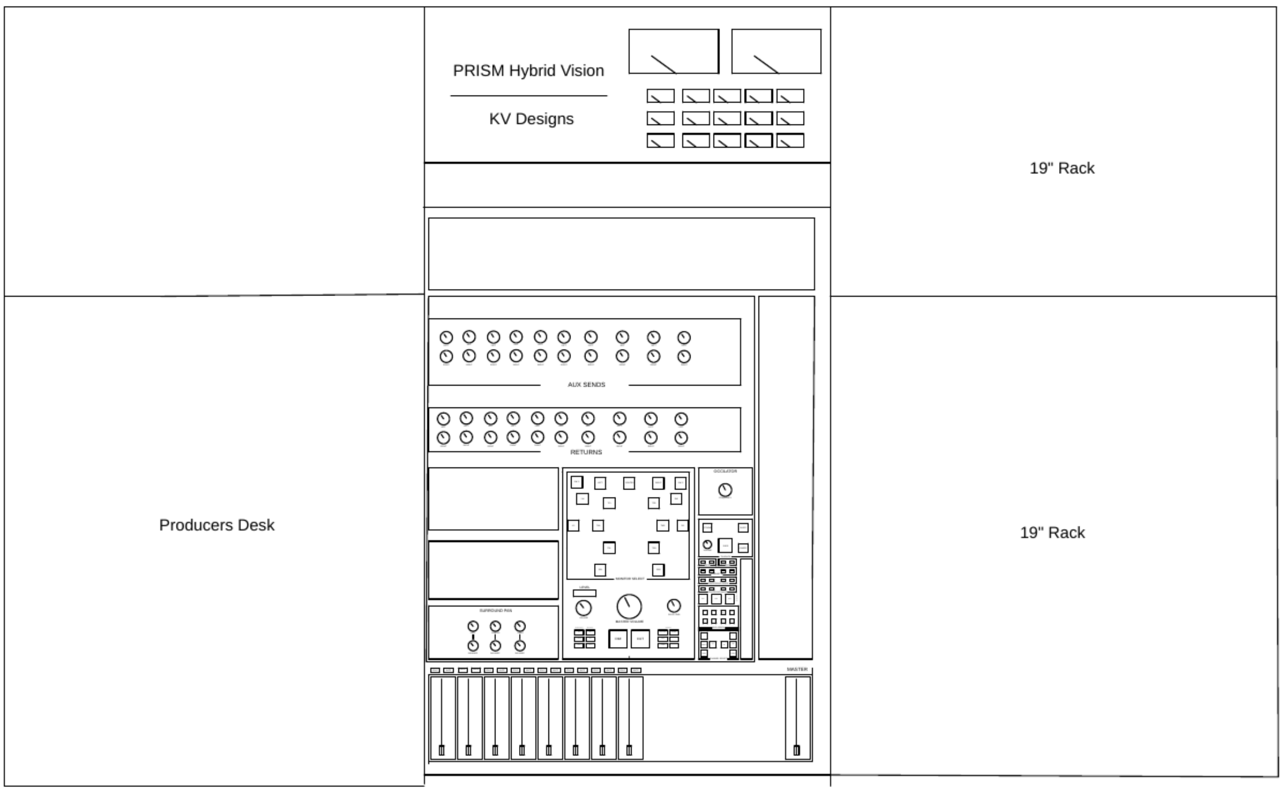
All modules are connected via a Samtec Tiger Eye™ TFM 60 pin TFM-130-22-S-D-A connector. The module standard is called uAMX (Universal Audio Module eXchange). This is so for future expansion, engineers can replace their modules with better ones instead of replacing the entire console.

Digital Control & Recall

- Microcontrollers: STElectronics NUCLEO
- Motorized Faders
- Digital Control via a computer connected to USB of the console
- Recall
- Presets

All Automation is controlled via an external computer connected to the internal computer and microcontroller which controls all presets, recall, automation, etc.

Center Console



Channel Control, Routing & Outputs

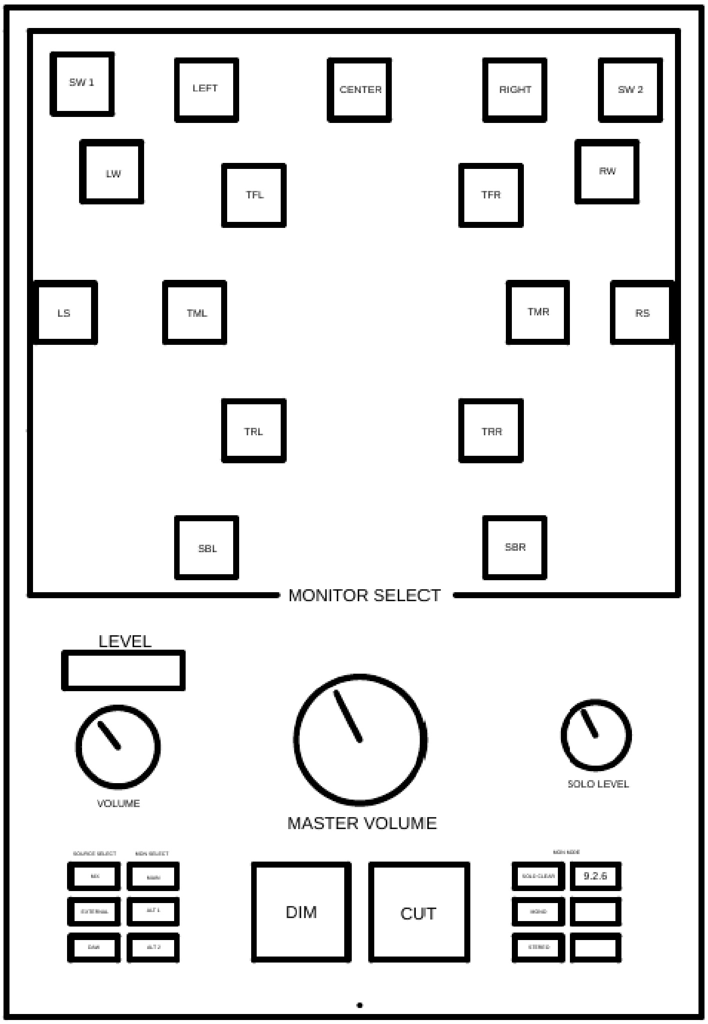
- Gain Staging: Hardware-Inverting active polarity reversal switch. 10kΩ Fine-Trim control pot with localized NE5532 unity buffer.

- Volume Fader: 100mm Bourns PSM01 high-torque motorized fader, 16-bit DC microcontroller positional feedback, and integrated capacitive touch-sensing array.
- AUX Matrix: 3 stereo AUX sends Pre/Post fader selection switches per bank.
- Output balancing section: Switchable styles of output balancing by OLD, NEW, and OLD NEW; utilizes the THAT 1646 for output balancing.
- Master Interface: DB25 Connectors to connect 8 channels to the center console at once.

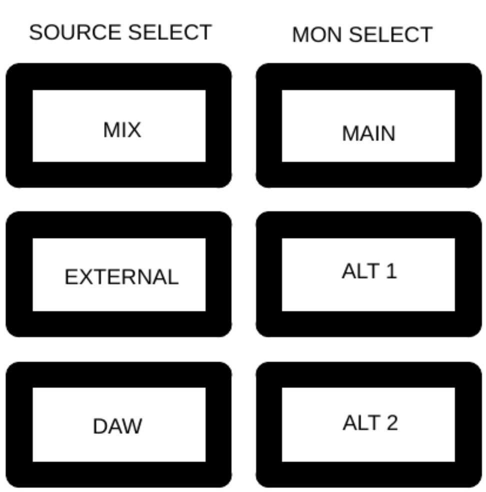
Spatial Mix & Automation

- Local Processing Sub-Groups: Per 8 channels, there is one STMicroelectronics NUCLEO board controller. This controller manages faders, channel buttons, etc.
- Immersive Spatial Panning: Managed by the ADAU1452 SigmaDSP Engine, This controller uses Vector based Amplitude Panning for spatial panning.
- User interface: Uses touchscreens in the center console for digital EQ and compressor plugins and control. Every 8 channels, there is one Screen to display groups and channel status.
- Outputs: 17 Monitor outpust for Dolby Atmos 9.2.6 Spatial Surround sound.

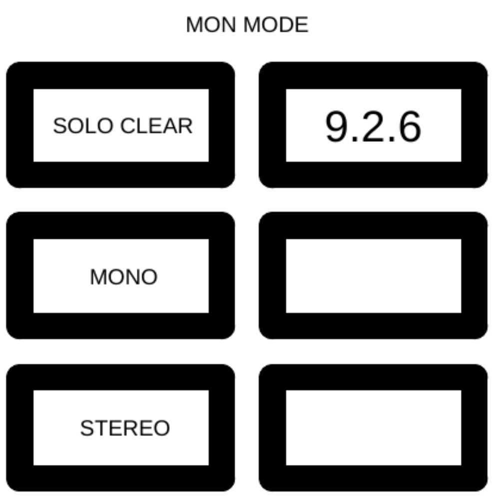
Monitor Controller



- Monitor select: This function selects the monitors by button. Such as: If your mix requires soloing a monitor you can select that monitor for the audio to play through or change the volume of the monitor.
- The master volume dial dims the master volume of the monitors, but does not change the main mix.
- The solo level changes how much of the solo channel's volume you want to hear.

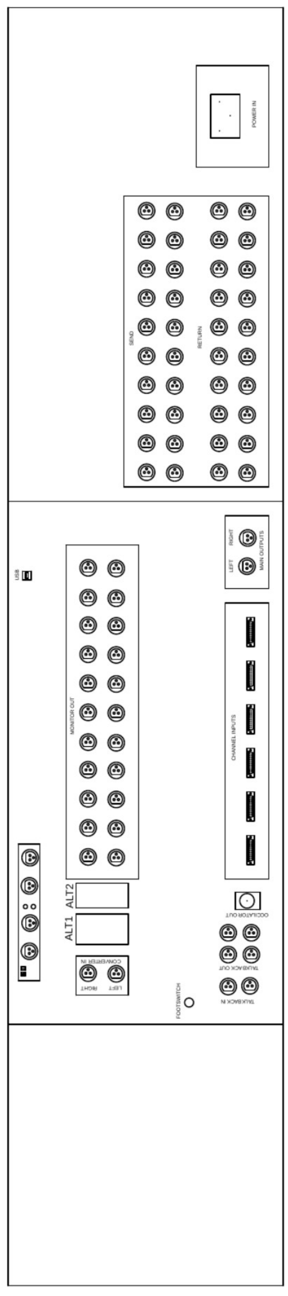


- Source Select: Source select changes the source to your monitors. However, this depends on your input to the monitor channels on your console (refer to Connectivity).



- Monitor select changes your output (This also depends on your connections to the console).
- Solo clear: Solo clear clears all solo channels you have currently soloed.
- Mono changes your output to a mono channel.
- Stereo changes your output to a stereo channel.
- 9.2.6 Selects all output monitors connected to 9.2.6 channels.

Connectivity



PRISM Console Connectivity

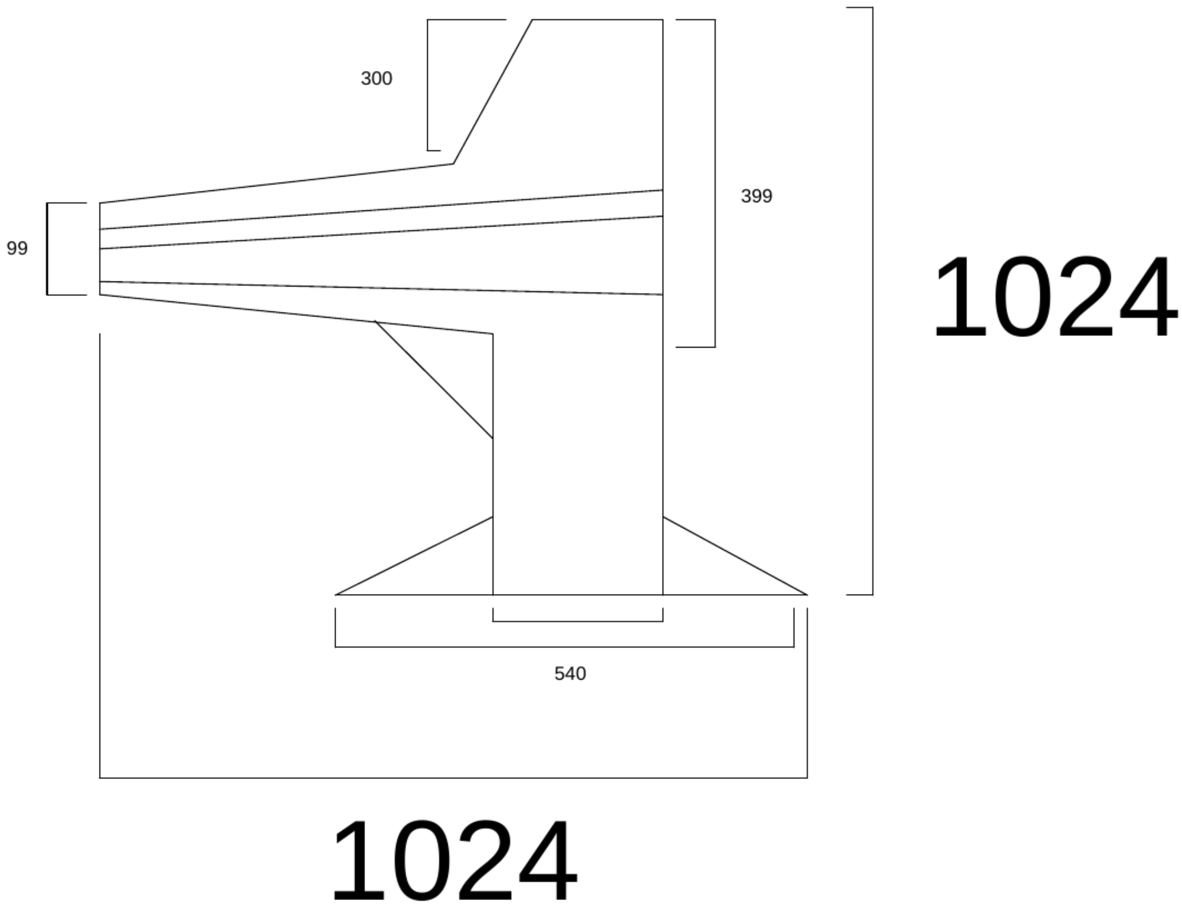
- 17x Monitoring Outputs
- 2x Stereo Monitor in
- 20x AUX Sends
- 20x AUX Returns
- 2x Mix Out
- 6x DB25
- 1x BNC
- 4X XLR Talkback Out
- 2x XLR Talkback In
- 1x Footswitch

RME ADI-2 PRO EX AD/DA Converter

- 2x4 channels
- 2x XLR-1/4” Analoge IN
- 2x XLR, 2x ¼” (main out) 2x ¼” (headphones) Analoge OUT
- 1x Optical (S/PDIF, ADAT) 1x DB-9 (AES/EBU, S/PDIF) Digital IN
- 1x Optical (S/PDIF, ADT) Digital Out
- 1x Locking USB-C

- 1x 1/8” TS

Dimensions*



Unit of measurement: mm

*Measurement NOT Final. When components are acquired; Components will me measured and console dimensions will be determined.